# Computational Studies in Identifying Potential Inhibitors of Pantothenate Synthetase including Machine learning and Molecular Docking approaches

*Dissertation submitted in fulfilment of the requirements for the Degree*

*of*

## MASTER OF TECHNOLOGY

By

**AGRANI SINHA**

Department of Biotechnology

**JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY**

(Declared Deemed to be University U/S 3 of UGC Act)

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# Computational Studies in Identifying Potential Inhibitors of Pantothenate Synthetase including Machine learning and Molecular Docking approaches

DISSERTATION

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IN PARTIAL FULFILLMENT FOR THE DEGREE OF

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**IN**

**BIOTECHNOLOGY**

By

Ms. Agrani Sinha

Under the supervision of

Dr. Imran Siddiqi
Senior Principal scientist

Division of Molecular and Structural Biology
CSIR -Central Drug Research Institute
Lucknow-226031

# TABLE OF CONTENTS


| Chapter No. | Topics | Page No. |
|---|---|---|
| | Declaration | 3 |
| | Acknowledgement | 4 |
| | Abstract | 7 |
| | List of figures | 8 |
| | List of Tables | 9 |
| | List of Abbreviations | 10 |
| Chapter-1 | Introduction | 12 |
| Chapter-2 | Review of literature | 20 |
| Chapter-3 | Materials and methods | 34 |
| Chapter-4 | Results | 42 |
| | References | 50 |

# DECLARATION

I hereby declare that the project work entitled "**Computational Studies in Identifying potential inhibitors of Pantothenate Synthetase including Machine learning and Molecular docking approaches**" was carried by me under the supervision of **Dr. Imran Siddiqi,** Senior Principal Scientist, Division of Molecular and Structural Biology, CSIR- Central Drug Research Institute, Lucknow. I have not submitted the matter embodied in this project report for the award of any other degree or diploma of any other university or institute.

Agrani Sinha,
M.Tech.
Biotechnology, JIIT,
Sector-62, Noida.
(U.P.)

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# ABSTRACT

# ABSTRACT

Health concern is our top priority and in today's era health sectors have growing economy due to the fact that many individuals are suffering from poor health. Tuberculosis is one of the top 10 leading disease even before HIV/AIDS. It is a disease in which extensive research is ongoing as it is increasing at higher rates due to developing of multidrug-resistant strains in an infected individual. Computational approach in finding the inhibitors of these proteins is a major challenge. The main target is focused on the survival of the bacteria. Therefore, selected the protein *Pantothenate Synthetase* produced from the gene name panC which mainly helps in the survival of the *Mycobacterium tuberculosis* bacterium. Designed the experiment to find the potential inhibitors of the mentioned protein using machine learning methods by training the models prepared of inhibitors and non-inhibitors dataset. Obtained the better accuracy in Support Vector Machine (SVM) by evaluating the confusion matrix through Matthews Correlation Coefficient (MCC) in MATLAB environment. The best result in docking of the molecules by the top 10 hits was obtained through Maybridge database. One can get much better accuracy in future while evaluating the model for more better outcome of the potential inhibitors. Further it can be used for in vivo inhibition assay and used for making a potential drug molecule.

**KEYWORDS*: Mycobacterium tuberculosis, Pantothenate Synthetase, Support Vector Machine, Computational approach, Drug Designing, Machine learning***

# LIST OF FIGURES


| Sr. No. | Figures | Page No. |
|---|---|---|
| 1. | Transmission of Tuberculosis | 13 |
| 2. | HIV-TB coinfection | 14 |
| 3. | Worldwide spread of tuberculosis (2018) | 15 |
| 4. | Symptoms of Tuberculosis. | 16 |
| 5. | Diagnosis through Tuberculin Skin test for tuberculosis. | 17 |
| 6. | Timeline History of Tuberculosis. | 20 |
| 7. | Latent TB infection Treatment Regimens. | 21 |
| 8. | Spectrum of *M.tuberculosis* Physiology. | 22 |
| 9. | TB regimens for Drug-susceptible TB | 22 |
| 10. | Multi-drug Resistant Tuberculosis and second line treatment. | 23 |
| 11. | Tuberculosis Stages of Infection | 24 |
| 12. | Open source Computer-aided drug designing | 25 |
| 13. | Homology Modelling Steps. | 27 |
| 14. | Molecular docking sequential major steps. | 28 |
| 15. | Machine learning | 29 |
| 16. | Application of Machine learning in drug discovery and development. | 30 |
| 17. | Machine learning workflow | 30 |
| 18. | Support Vector machine outlay | 31 |
| 19. | Reaction mechanism of Pantothenate Synthetase | 35 |
| 20. | Flowchart describing the process of formation of Pantothenate. | 36 |
| 21. | Data Pre-processing. | 37 |
| 22. | MATLAB Working Interface displaying the training of Protein Model. | 38 |
| 23. | Molecular docking | 40 |
| 24. | Flowchart of the process. | 40 |
| 25. | Confusion matrix of Obtained Model | 43 |
| 26. | Protein Visualized in UCSF Chimera. | 44 |
| 27. | Ligand as Nafronyl Oxalate. | 44 |
| 28. | Ligand-Protein Interaction of control. | 46 |
| 29. | Ligand interaction of 10 compounds | 46 |

# LIST OF TABLES


| Sr.no. | Tables | Page no. |
|---|---|---|
| 1. | Homology Modelling Steps. | 26 |
| 2. | Model data | 43 |
| 3. | Ligand interaction and docking score of control and 10 compounds. | 45 |

# LIST OF ABBREVIATIONS

| | |
|---|---|
| TB | Tuberculosis |
| MCC | Matthews Correlation Coefficient |
| SVM | Support Vector Machine |
| Mtb | Mycobacterium Tuberculosis |
| Bi | Bimolecular |
| Pan | Pantothenate |
| nM | Nanomolar |
| .smi | SMILE |
| .csv | Comma Separated values |
| HIV | Human Immunodeficiency Virus |
| AIDS | Acquired Immunodeficiency Syndrome |
| DUD-E | Directory of Useful Decoys- Extended |
| PDB | Protein Data Bank |
| SDF | Spatial Data File |
| UniProt | Universal Protein Resource |
| UniProtKB | UniProt Knowledge Base |

# CHAPTER 1

# INTRODUCTION

## 1.1 Infectious Diseases:

Type of disorders caused by organisms is termed as infectious disease. The organisms which causes the infections are Bacteria, Viruses, Fungi or Parasites. They live inside or outside of the living entity. They are normally not harmless and even helpful but under certain conditions may cause disease. The infectious diseases will be passed from person to person or through contaminated food or water or getting transmitted through insects or other animals. It often includes fever and fatigue as signs of symptoms and mild infection needs rest or home remedies while in critical condition needs hospitalization. The causes from different type of organisms vary depending upon the site of infection such as Bacteria are one-cell organisms and are chargeable for illnesses like tract infections and Tuberculosis. Viruses are even smaller than bacteria and may cause severe respiratory illness to AIDS. Many Fungi cause skin diseases like ringworm and plenty of other affects lungs or system. The parasites like mosquitoes causes malaria and many other parasites may be transmitted through contaminated food and water. [1]

The prevention to these diseases is frequently washing of hands, avoiding close contact, getting vaccinated, staying home when ill, preparing food safely, travelling safely and many other measures taken as much as an individual can adapt such habits.

## 1.2 Tuberculosis:

It is mainly caused by the pathogenic bacteria *Mycobacterium tuberculosis* which mainly attacks the lungs and settles in that area and grow, from there they spread to other parts of the body through blood like Kidney, Brain or maybe Spine. Mycobacteria is neither gram positive or gram negative and has aberrant waxy coating on its cell surface due to the vicinity of mycolic acid, this coating makes them of acid-fast bacteria with poor absorption and high retention. It is highly aerobic and requires high level of oxygen and divides every 15-20 hours. It is infected to an individual through the spread of TB bacteria through air by getting in close contact to an infected individual when they speak, laugh or frequently cough, the other individual may breathe in the bacteria and get infected. [2]

Vaccines for Tuberculosis is Bacille Calmette-Guérin (BCG) vaccine. It is delivered to infants and small children through injections in countries where TB is common. It is given to the patients who are immune to the isoniazid and rifampin.

TB is divided into two forms based on their mode of activation: Latent TB and Active TB.

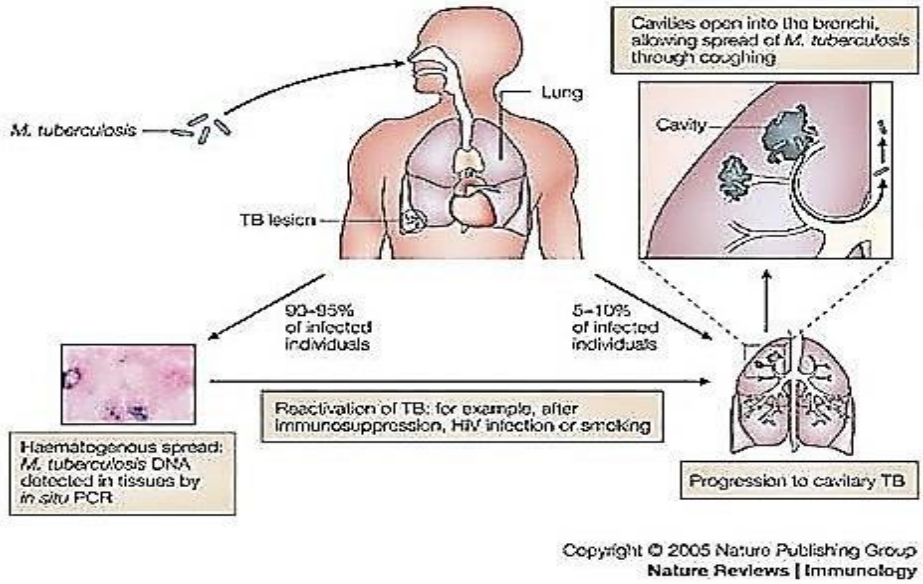



**Figure 1.1.** Transmission of tuberculosis.

### 1.2.1. HIV-TB coinfection:

Tuberculosis with HIV coinfection is a serious health concern. Over the world maximum number of deaths with HIV persist with patients suffering from tuberculosis. If left untreated TB disease can progress from sickness to death as the immune system of an individual is weakened. Among the infected population, HIV Infected individual are more prone to develop TB disease. AIDS defining condition also persists. Role of Immune system in the progression of HIV/TB coinfection takes place. Reduced level of CD4+ cells results in the progression of AIDS and CD8+ cells controls the growth of latent TB infection. The tumour necrosis factor (TNF -α) mediated macrophage apoptotic response is also impaired by HIV which leads to the increased bacterial survival. The CD4+ and TNF plays an important role in Granuloma formation and it decreases in response to growth of infection.

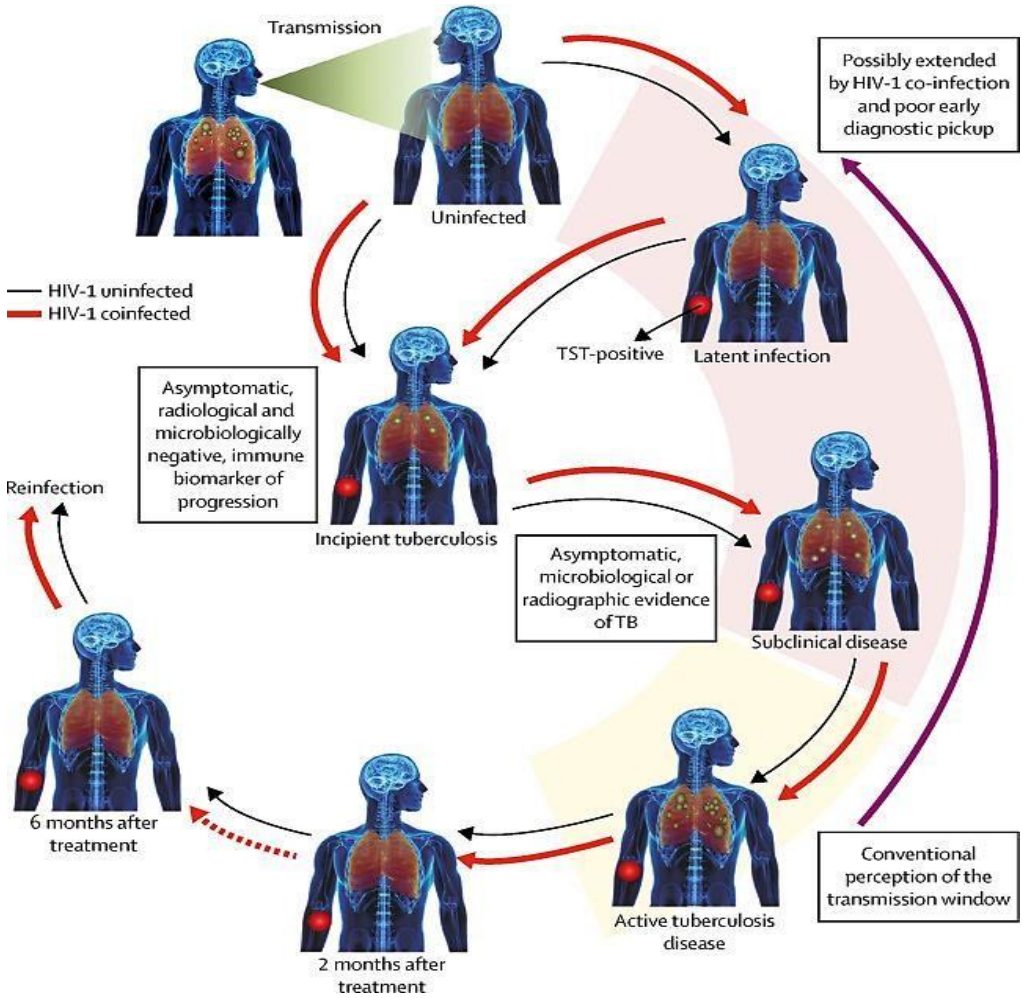



**Figure 1.2.** HIV-TB coinfection.

### 1.2.2. Tuberculosis Estimate around the world:

Around the globe, approximately 1.4 million lives are at risk every year. The main reason is the growth of multi-drug resistant strains of bacterium adapted with immune responses within the host system due to which many people die each year at an alarming rate. [3]

The goal for TB control worldwide is set to target as active TB should be less than one case per million population per year by 2050 [3].

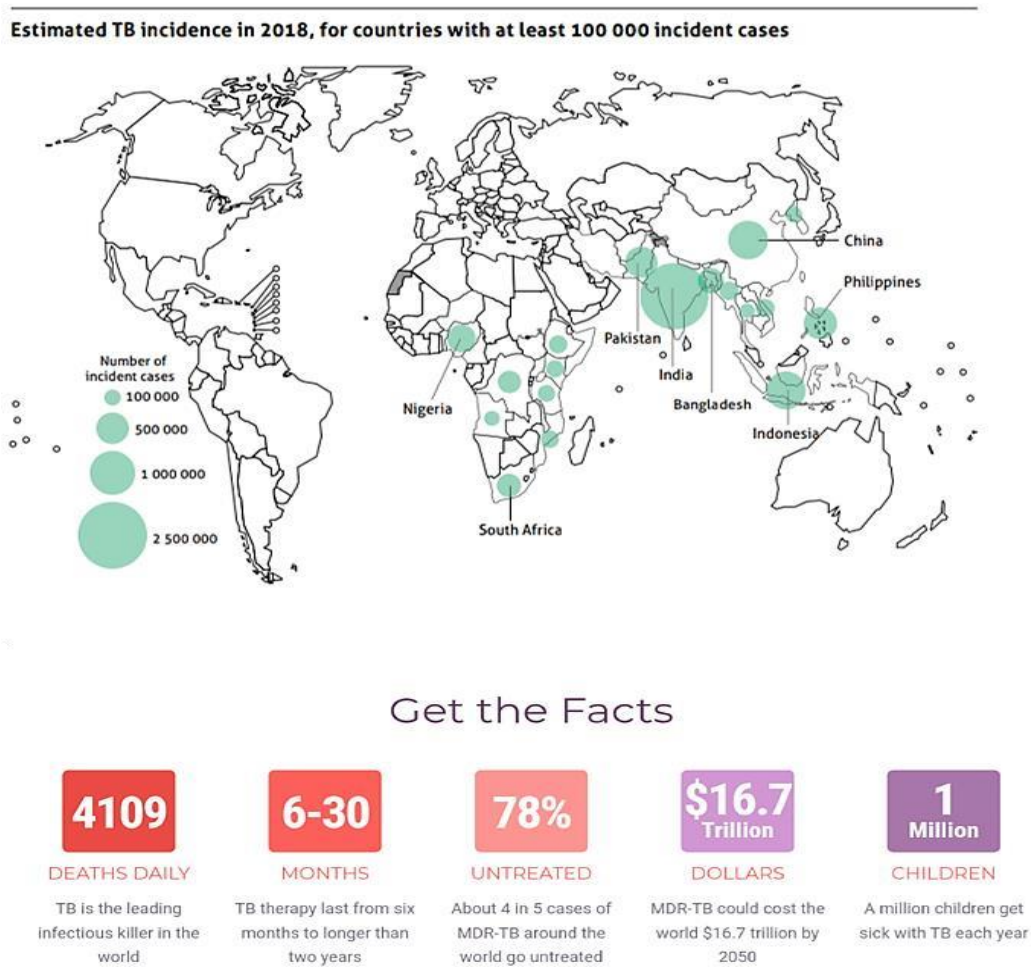



**Figure 1.3.** Worldwide spread of tuberculosis (2018).

### 1.2.3. Symptoms and diagnosis of Tuberculosis:

Population with weak immunity can get TB easily. The symptoms of Tuberculosis arise from mainly two conditions: **Latent TB and Active TB**. In Latent TB the patient immune system fights with the germs and no symptoms takes place but if it persists for a longer duration than active TB may take place. In active TB germs may multiply and can make you sick which includes cough with high fever in a long duration of time with sudden weight loss, night sweats and loss of appetite therefore resulting in weakening of immune system in a host body. Treatment is done in 6-12 months duration of time.

The Diagnosis of tuberculosis includes **Skin test** and **Blood Test**. In skin test which is also known as **Mantoux Tuberculin skin test** includes the injection of small amount of fluid into the skin of lower arm. After 2-3 days checking of swelling determines the result. The result is predicted in three ways – *Positive, False Positive and False Negative*. In Positive result the individual is infected with TB bacteria. In False Positive, the test will predict you may have TB but in real you do not have and will be treated with Bacillus Calmette – Guerin (BCG) vaccine. In False negative, the individual does not have TB but the infection is recent. [4]

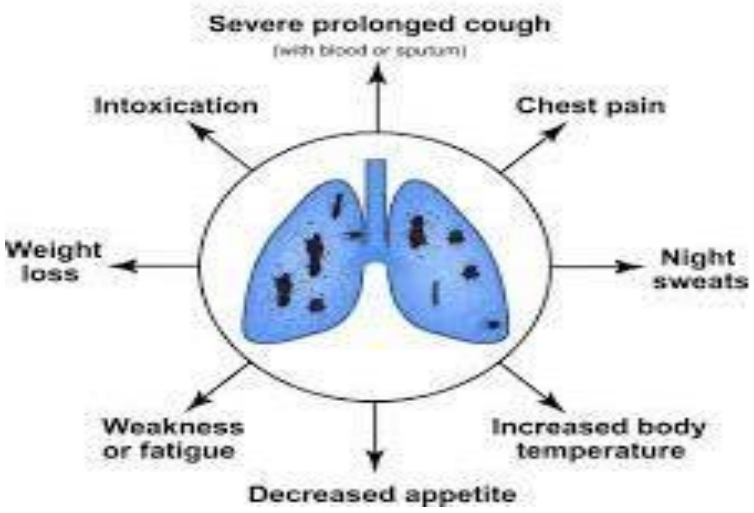



**Figure 1.4.** Symptoms of Tuberculosis.

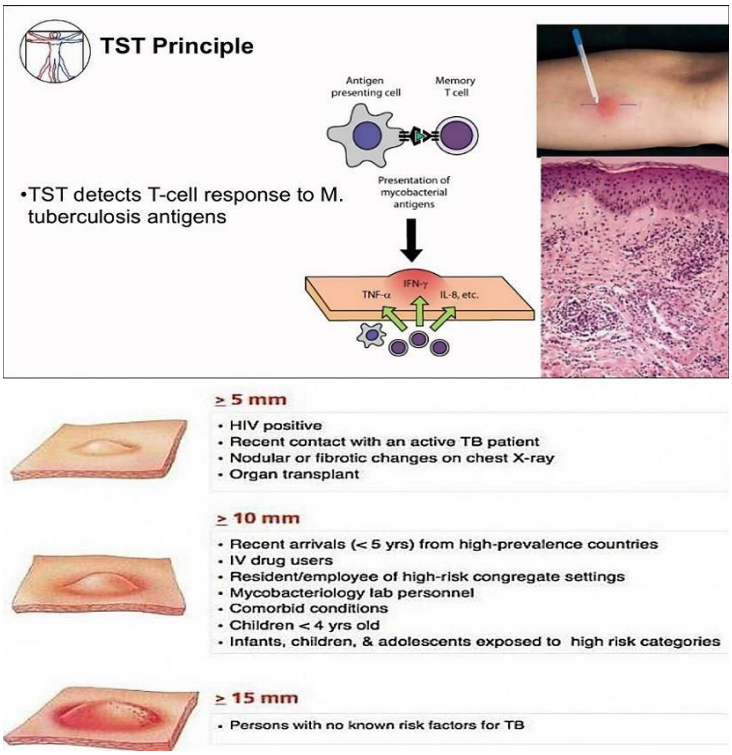



**Figure 1.3.** Diagnosis through Tuberculin Skin test for tuberculosis.

## 1.3. Aims and Objectives:

**Aim:** To find the potential inhibitor of Pantothenate synthetase protein found in *Mycobacterium Tuberculosis.*

**Objective:** Identified the potential inhibitors of Pantothenate Synthetase from computational approach through the following procedure:

- Machine learning in MATLAB environment:
    - Data collection
    - Data Preparation
    - Model Generation (SVM)
- Virtual screening from Maybridge Compound Database
- Molecular docking in Schrodinger Maestro

## 1.5. Format of Thesis:

1. Introduction
    - 1.1 Infectious Diseases
    - 1.2 Tuberculosis
        - 1.2.1. HIV-TB coinfection
        - 1.2.2. Tuberculosis Estimate around the world
        - 1.2.3. Symptoms and diagnosis of Tuberculosis
    - 1.3 Aims and Objectives.
2. Review of literature
    - 2.1 History
    - 2.2 Drugs used for treatment
        - 2.2.1. Treatment for Latent TB infection
        - 2.2.2 Treatment for TB Disease
        - 2.2.3 Multidrug Resistance
    - 2.3 Tuberculosis Lifecycle
    - 2.4 Computer Aided Drug- Design (CADD)
    - 2.5 Drug Design
        - 2.5.1. Ligand-based drug design
        - 2.5.2. Structure-based drug design
    - 2.6. Homology Modelling
    - 2.7. Molecular Docking
    - 2.8. Machine learning Based Drug design
        - 2.8.1. Supervised learning
        - 2.8.2. Unsupervised learning
        - 2.8.3. Support Vector Machines (SVM)
        - 2.8.4. MATLAB Interface.
3. Materials and Methods
    - 3.1 Essentiality of Potential Drug target
    - 3.2. Pantothenate Synthetase as a potential target
    - 3.3. Computational approach in finding the leads
    - 3.4. Protein and Inhibitor Database
    - 3.5. Data Pre-Processing
    - 3.6. Descriptor Calculation
    - 3.7. Model Building
    - 3.8. Virtual Screening
    - 3.9 Molecular docking
    - 3.10 Flowchart of the process.
4. Result
    - 4.1. Database collection and refinement
    - 4.2 Model Building
    - 4.3. Molecular Docking
        - 4.3.1. Ligand preparation
        - 4.3.2 Grid Preparation
    - 4.4. Discussion
    - 4.5. Conclusion

# CHAPTER 2

# REVIEW OF LITERATURE

## 2.1 History:

Tuberculosis is the most prevailing infectious disease as it is a rapidly occurring air-borne infection causing the damage to respiratory system in humans. It is mainly caused by *Mycobacterium tuberculosis* (Mtb) which may be a slow growing bacterium which thrives for living under the host rich in blood and oxygen as in lungs. Mycobacterium was discovered by **Dr. Koch** on March 24, 1882, during that time it killed one out of each seven people U.S and Europe and it was known to be as "The great white plague". Today March 24 is designated as World TB day to educate people and make them aware of the global crisis of TB. In the past 15,000 years it has been present as suggested by the genetic studies. "Tuberculosis" term was coined in 1834 by **Johann Schonlein**. In 1700, due to the paleness of the patients, TB was called as "White plague" and in 1800 it was also called "Consumption". For Diagnosis of tuberculosis came around in 19$^{th}$ century was "Tuberculin Skin test" developed by **Clemens von Pirquet** in 1907 and after 3 years came into existence as to demonstrate latent tuberculosis infection in asymptomatic children. The discovery of its bacterial cause made it to combat the spread of tuberculosis around 19$^{th}$ century. During World war I, BCG vaccine was widely employed. In the modern times, Tuberculosis control and treatment was indicated by the discovery of Streptomycin and Isoniazid in 1944 and 1952. Today, Treatment for tuberculosis is not a major challenge due to the extensive research ongoing but also a difficult task to make its progress a limiting factor.[7][8]

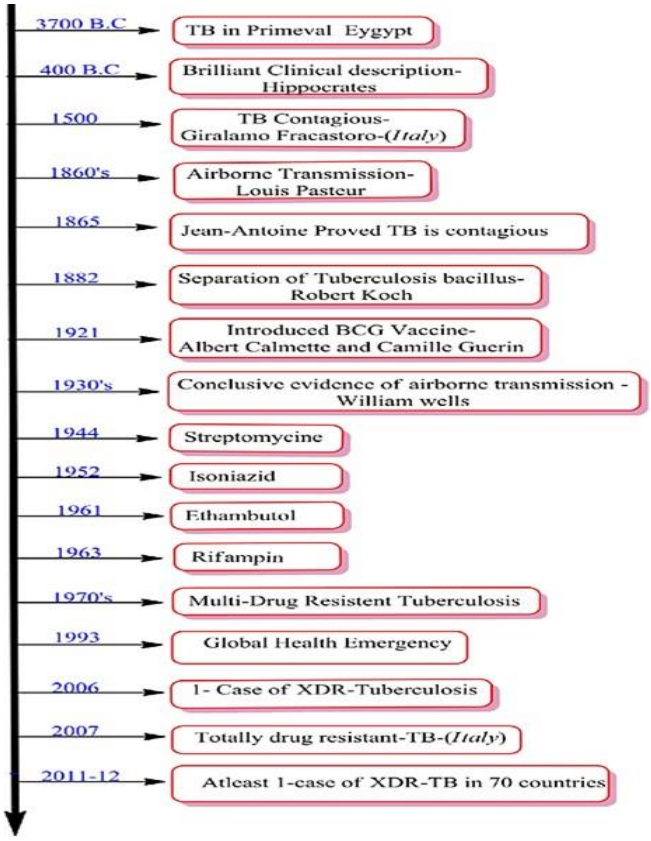



**Figure 2.1.** Timeline History of Tuberculosis.

## 2.2 Drugs used for treatment:

The treatment should be started as early as possible to limit the progress of latent TB to an active TB infection. If the symptoms start occurring, one should get their diagnosis. If the diagnosis results are positive, they should further go for the treatment. [9]

### 2.2.1. Treatment for Latent TB infection:

The medications used for Latent TB infection include the first-line drugs: Isoniazid (INH), Rifapentine (RPT) and Rifampin (RIF). These medications can be taken in longer durations or in combinations. [30]

| Drug(s) | Duration | Dose | Frequency | Total Doses |
|---|---|---|---|---|
| Isoniazid (INH)* and Rifapentine (RPT)' | 3 months | Adults and Children aged 12 years and older:<br>INH: 15 mg/kg rounded up to the nearest 50 or 100 mg; 900 mg maximum<br>RPT:<br>10–14.0 kg 300 mg<br>14.1–25.0 kg 450 mg<br>25.1–32.0 kg 600 mg<br>32.1–49.9 kg 750 mg<br>≥50.0 kg 900 mg maximum<br>Children aged 2–11 years:<br>INH*: 25 mg/kg; 900 mg maximum<br>RPT': as above | Once weekly | 12 |
| Rifampin (RIF)[§] | 4 months | Adults: 10 mg/kg<br>Children: 15–20 mg/kg[‖]<br>Maximum dose: 600 mg | Daily | 120 |
| Isoniazid (INH)* and Rifampin)[§] | 3 months | Adults:<br>INH*: 5 mg/kg; 300 mg maximum<br>RIF[§]: 10 mg/kg; 600 mg maximum<br>Children:<br>INH*: 10-20 mg/kg; 300 mg maximum<br>RIF[§]: 15-20 mg/kg; 600 mg maximum | Daily | 90 |
| Isoniazid (INH) | 6 months | Adults: 5 mg/kg<br>Children: 10–20 mg/kg[¶]<br>Maximum dose: 300 mg | Daily | 180 |
| | | Adults:15 mg/kg<br>Children: 20–40 mg/kg[¶]<br>Maximum dose: 900 mg | Twice weekly[‡] | 52 |
| | 9 | Adults: 5 mg/kg<br>Children: 10–20 mg/kg[¶]<br>Maximum dose: 300 mg | Daily | 270 |

**Figure 2.1.** Latent TB infection Treatment Regimens.

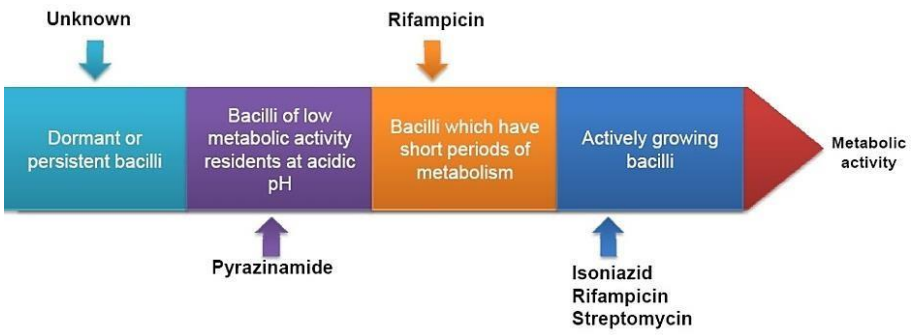



**Figure 2.2.** Spectrum of *M.tuberculosis* Physiology.

### 2.2.2. Treatment for TB Disease:

TB disease will be treated by taking the prescription drugs for 6-9 months. Currently approved First-line drugs by the U.S. Food and Drug administration (FDA) are: Isoniazid (INH), Rifampin (RIF), Ethambutol (EMB), Pyrazinamide (PZA). TB regimens for drug susceptible TB treatment have an intensive phase of 2 months and addition of continuation phase of 4 to 7 months in total 6-9 months. [32]

| | INTENSIVE PHASE | | CONTINUATION PHASE | | | | |
|---|---|---|---|---|---|---|---|
| Regimen | Drugs[a] | Interval and Dose[b] (minimum duration) | Drugs | Interval and Dose[b,c] (minimum duration) | Range of Total Doses | Comments[c, d] | Regimen Effectiveness |
| 1 | INH RIF PZA EMB | 7 days/week for 56 doses (8 weeks) *or* 5 days/week for 40 doses (8 weeks) | INH RIF | 7 days/week for 126 doses (18 weeks) *or* 5 days/week for 90 doses (18 weeks) | 182 to 130 | This is the preferred regimen for patients with newly diagnosed pulmonary TB. | Greater |
| 2 | INH RIF PZA EMB | 7 days/week for 56 doses (8 weeks) *or* 5 days/week | INH RIF | 3 times weekly for 54 doses (18 weeks) | 110 to 94 | Preferred alternative regimen in situations in which more frequent DOT during continuation phase is difficult to achieve. | |
| 3 | INH RIF PZA EMB | 3 times weekly for 24 doses (8 weeks) | INH RIF | 3 times weekly for 54 doses (18 weeks) | 78 | Use regimen with caution in patients with HIV and/or cavitary disease. Missed doses can lead to treatment failure, relapse, and acquired drug resistance. | Lesser |
| 4 | INH RIF PZA EMB | 7 days/week for 14 doses then twice weekly for 12 doses[e] | INH RIF | Twice weekly for 36 doses (18 weeks) | 62 | Do not use twice-weekly regimens in HIV-infected patients or patients with smear positive and/or cavitary disease. If doses are missed then therapy is equivalent to once weekly, which is inferior. | |

**Figure 2.3.** TB regimens for Drug-susceptible TB.

### 2.2.3. Multidrug Resistance:

MDR TB is caused by bacteria that is resistant against a minimum of two potent drugs isoniazid (INH) and rifampin (RIF). Extensively drug-resistant TB (XDR-TB) could be a kind of MDR which is rare which is immune to Isoniazid, rifampin additionally fluoroquinolone and a minimum of 3 injectable second-line drugs. It is a major concern worldwide and need progressive research to bring the best outcome for its treatment as much as possible. [10]

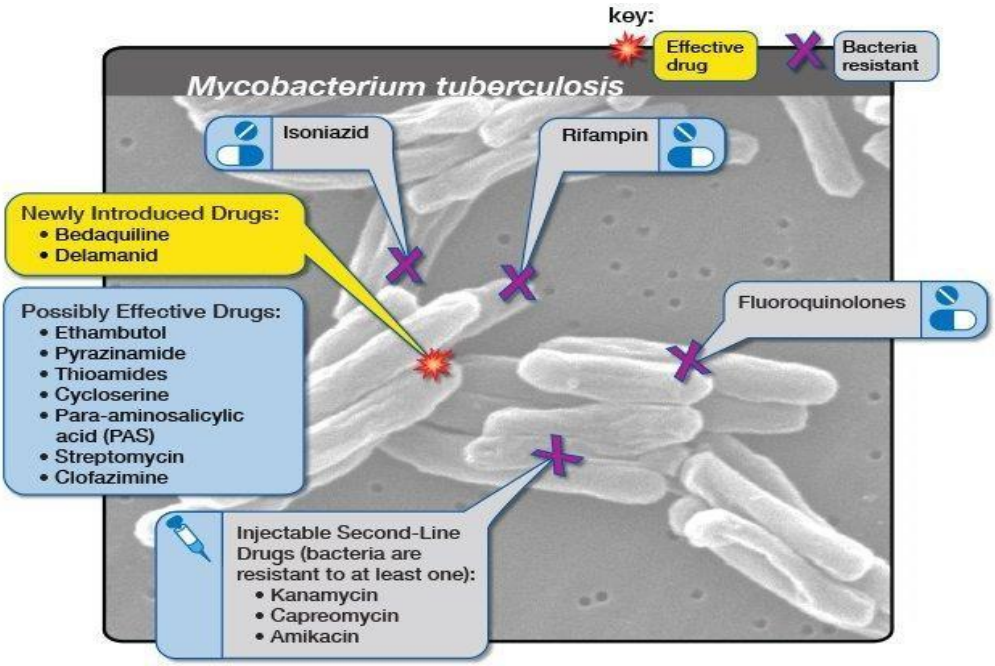



**Figure 2.4.** Multi-drug Resistant Tuberculosis and second line treatment.

## 2.3 Tuberculosis Lifecycle:

TB infects the body in five consecutive phases: The initial macrophage response, Growth stage, immune control stage and the lung cavitation stage. These Phases happen within one month.
**Stage One** is the initial stage of the infection in which the tuberculosis bacteria enters the lungs and alveolar macrophages of the host system through air. This bacterium than initiate to reproduce and infect.
**Stage two** is the stage of symbiosis in which bacteria reproduces for at least 7-21 days if not killed.
**Stage three** initiates the stage of caseous necrosis lasting for 14-21 days. In this stage, tuberculosis begins to develop where their binary fission is decreased and they kill non-activated macrophages around them. These leads to the anoxic conditions and lowering of pH simultaneously.
**Stage four** response comes into action after 21 days of infection where tissue damaging and macrophage activation takes place. The spread of tubercle increases throughout the lungs where mycobacterium tuberculosis uses non activated macrophages to reproduce which causes tubercle to grow. It can also spread into the bloodstream and patient can develop tuberculosis outside the lungs.
**Stage five** is liquefication and cavity formation stage in which tubercle liquify and spread at faster rates. A small number of patients develop these types of cases. [11]

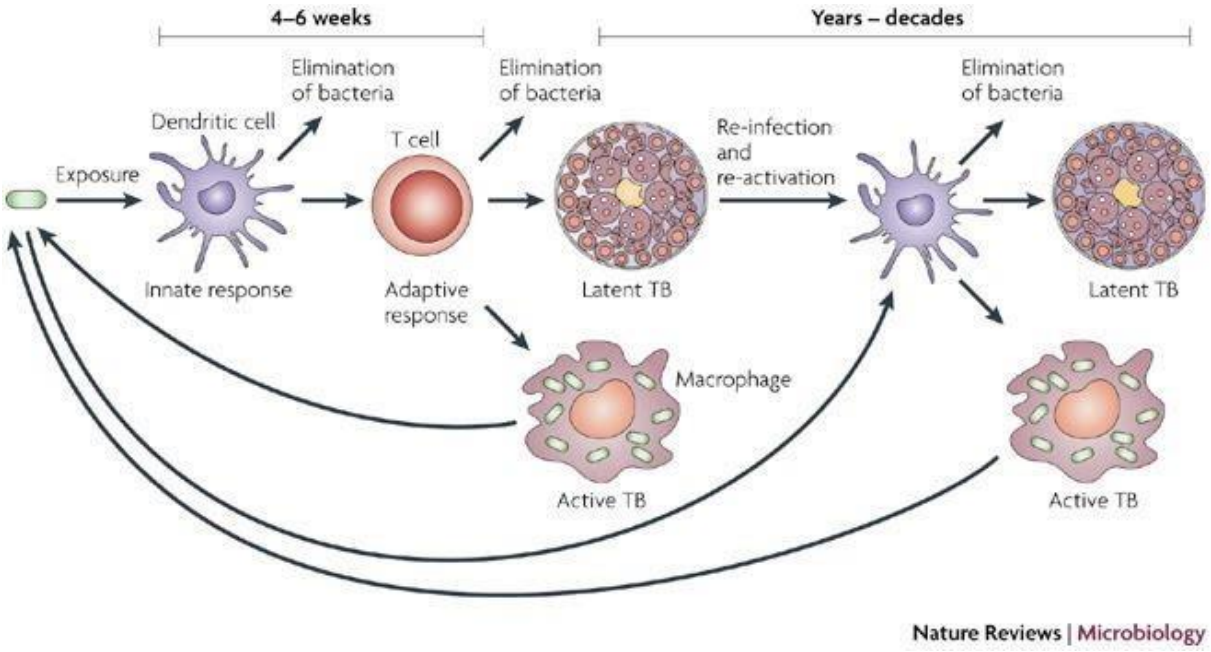



**Figure 2.5.** Tuberculosis Stages of Infection.

## 2.4. Computer Aided Drug- Design (CADD):

It makes you predict the compatibility of the drug molecule with the protein target. It includes molecular mechanics and molecular dynamics which is used to estimate their strength of intermolecular interactions.
Molecular mechanics methods provide semi-quantitative prediction of the binding of the molecules. These methods include Linear regression, Machine learning and neural nets or other statistical techniques. [12] [13]
The drug design may be used for following stages of drug discovery:

1) Virtual screening in Hit identification as in structure and ligand-based discovery
2) Affinity and selectivity using hit-to-lead optimization as in structure-based design, QSAR etc.
3) Maintaining affinity as in lead optimization of other pharmaceutical property.

In protein-ligand interaction, post-screening analysis has been developed with improved enrichment and effective mining potential candidates:

- Consensus scoring
- Cluster analysis

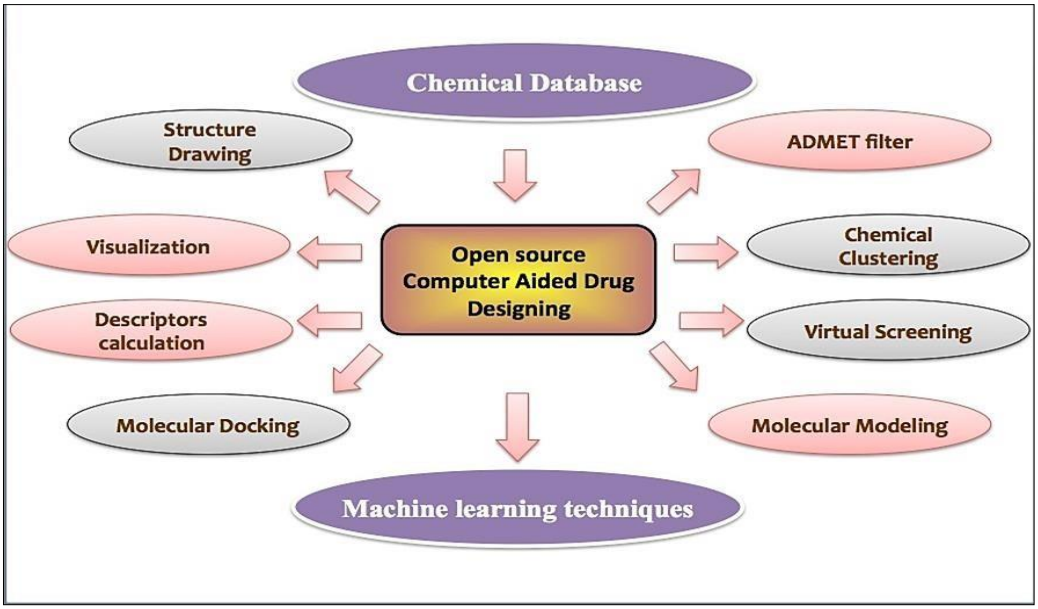



**Figure 2.6.** Open source Computer-aided drug designing.

Two major forms of Drug design include:

- Ligand-based drug design
- Structure-based drug design

### 2.4.1. Ligand-based drug design:

Also referred to as indirect drug design is predicated on the knowledge of other molecules that binds to the Biological Target of interest. These molecules are accustomed to derive the pharmacophore model that defines the structural characteristics so as to bind to the specified target. In contrast, **Quantitative-structure activity relationship (QSAR)** within which the relation between calculated properties and biological activity of the molecules is determined. These relationships successively predict the activity of the new analogues.[14]

### 2.4.2. Structure-based drug design:

Also referred to as direct drug design is predicated on the knowledge of the 3D structure of the biological target obtained through X-ray crystallography or NMR spectroscopy. If target is not available than homology modelling is completed to induce the required biological target. After getting the structure, the candidate drugs are predicted to bind with high affinity and selectivity to it.

The Structure-based drug design are roughly divided into three main categories:

1. Identification of the specified ligands from the massive chemical 3D database of small molecules to suit within the binding pocket of the biological target using fast docking methods. Method referred to as Virtual Screening.

1) De novo design of recent ligands. The ligand molecules are built up within the binding pockets of biological target by assembling the individual atoms or molecular fragments in stepwise manner.

2) Optimization of known ligands by evaluation of proposed analogs within the binding cavity.

## 2.6. Homology Modelling:

The procedure includes the four sequential steps:

- Template selection
- Target-template alignment
- Model construction
- Model assessment

| Sr.no. | Steps | Tools |
|---|---|---|
| 1. | Select Protein sequence and family | BLAST, Pfam |
| 2. | Select known X-ray Structures | PDB |
| 3. | Sequence alignment | FASTA, Clustal |
| 4. | Create Homology model Framework | Quanta |
| 5. | Introduce deletions, insertions | Quanta, PDB |
| 6. | Transfer new side chains to model | Quanta, Charmm |
| 7. | Introduce S-S, ion binding sites | Quanta, Charmm |
| 8. | Energy minimization | Charmm |
| 9. | Evaluate Model | Procheck |

**Table 2.1.** Homology Modelling Steps.

## 2.7. Molecular Docking:

Involves the bioinformatic modelling in which comprises the interaction of two or more molecules to give stable chemical compound. It predicts the 3D structure of ligand-target complex. [15]

Types of docking include:

- Flexible ligand docking target as rigid molecule
- Rigid body docking both target and ligand as rigid molecule
- Flexible docking both target and ligand as flexible molecule.

Applications of Molecular docking includes:

- Lead Optimization
- Hit identifications
- Drug- DNA interactions.

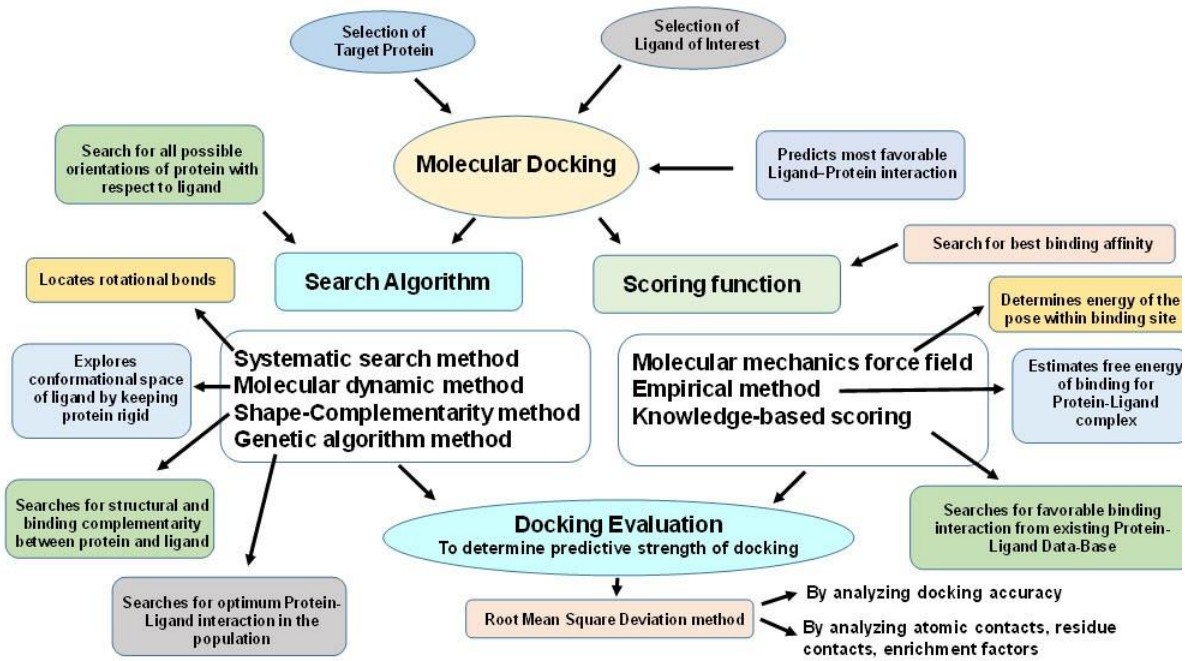



**Figure 2.7.** Molecular docking sequential major steps.

## 2.8. Machine learning Based Drug design:

Machine learning is an application of Artificial Intelligence, it is used to train the machine to access the data/info and make them learn for themselves through various specific algorithms in programming language. From the past, machine learning has given us Self-driving cars, effective web search and also improved understanding of the human genome. Working of machine learning relies on two techniques: **Supervised and Unsupervised Learning.**
In supervised learning, the model is trained with known input data to predict the output. In unsupervised learning, finding of intrinsic features and patterns of an input data is employed.

*2.8.1. Supervised learning:*

Based on the uncertainty of the data, Model is built to make accurate predictions. It uses Classification and regression techniques.

In classification models classify the input data to predict discrete response. It includes common algorithms: Support Vector Machines (SVM), Boosted and bagged Decision trees, k-nearest neighbours, Naive bayes, discriminant analysis, Logistic regression and neural networks.

In regression models, predict the continuoual responses of the input data. It includes algorithms such as: Linear model, Non-linear model, Regularization, Stepwise regression, Boosted and bagged Decision trees, Neural networks and adaptive neuro-fussy learning.

*2.8.2. Unsupervised learning:*

The input data does not require labelling and used to find the intrinsic structures within the data. Clustering is the common technique employed in unsupervised learning.

Common algorithms for performing clustering includes K-means, Hierarchical clustering, Hidden Markov models, self-organizing maps and subtractive clustering.

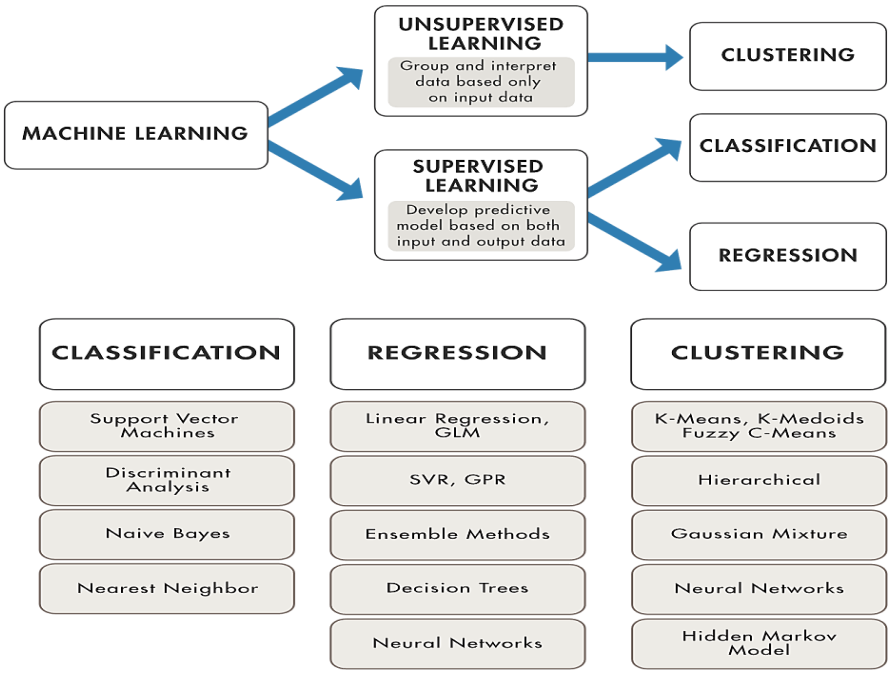



**Figure 2.8**. Machine learning.

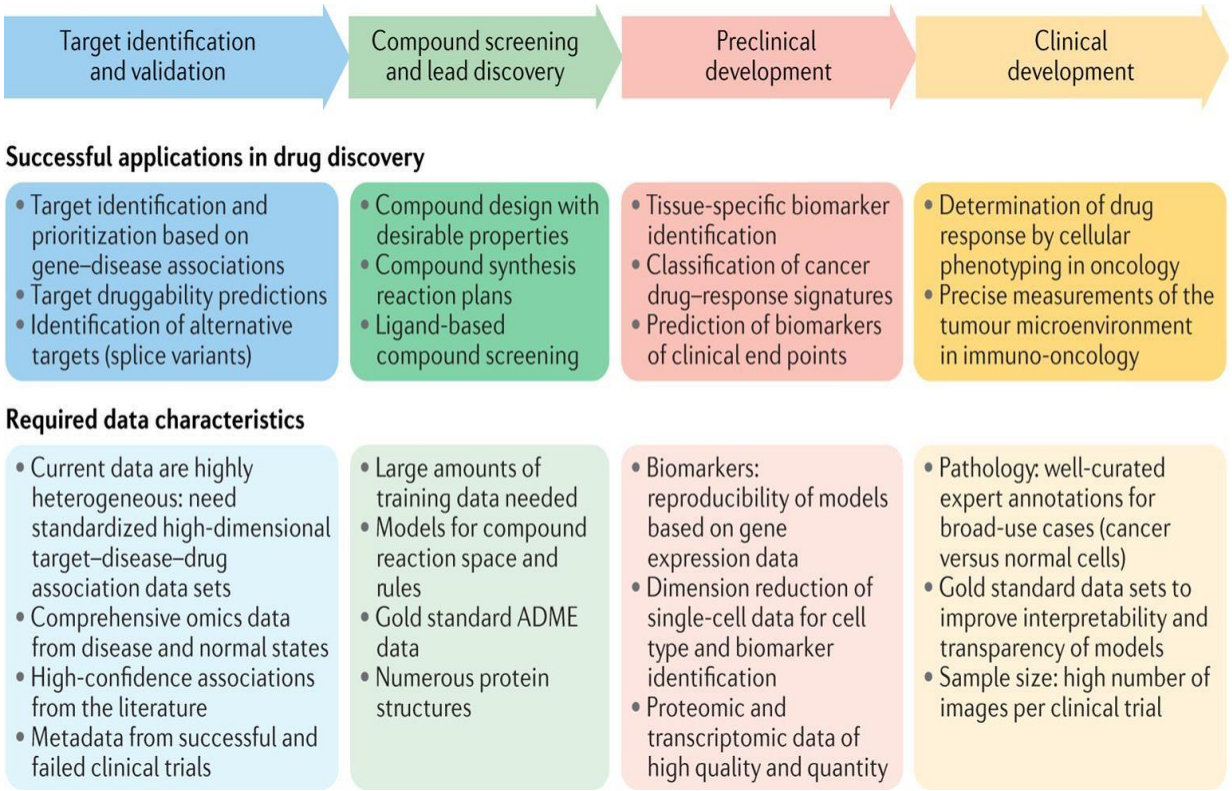



**Figure 2.9.** Application of Machine learning in drug discovery and development.

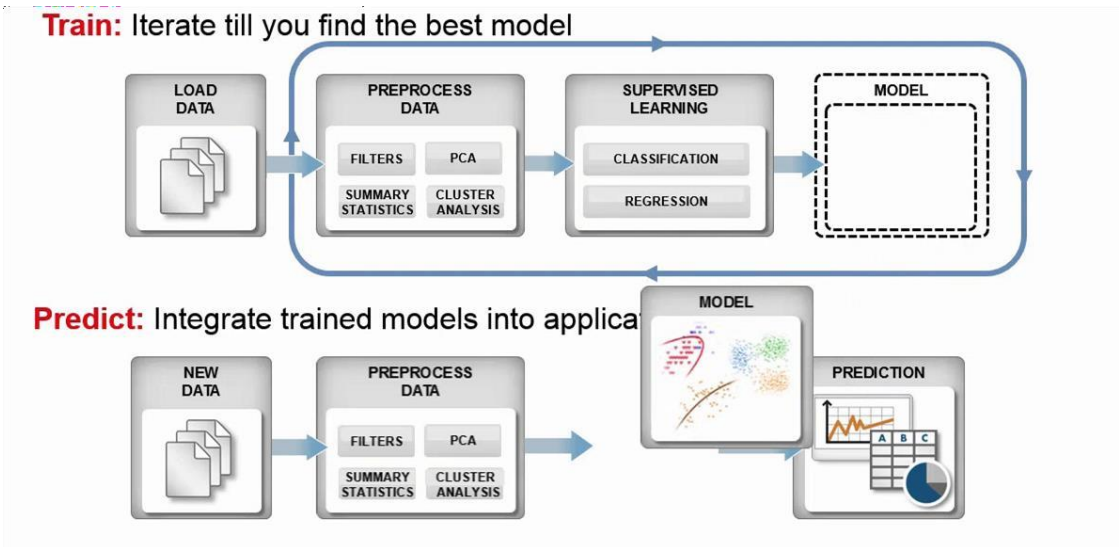



**Figure 2.3.1:** Machine learning workflow.

*2.8.3. Support Vector Machines (SVM):*

Support Vector machines is generally used for classification. SVM method is used for classification, regression and outliers detection. Support vectors are the coordinates of a particular observation. The plotting of each data point is done in n-dimensional space where n is the number of features. The hyperplane is built in such a manner to divide two classes very well. [42]

The following concepts in SVM are:

- **Support vectors:** The hyperplane defines such datapoints as support vectors.
- **Hyperplane:** It is basically a decision plane which classifies or divides the data points with different classes.
- **Margin:** The perpendicular distance between the hyperplane and data points or support vectors. If the distance is large than it is considered to be a good margin and if the distance is small than it is considered to be the bad margin.

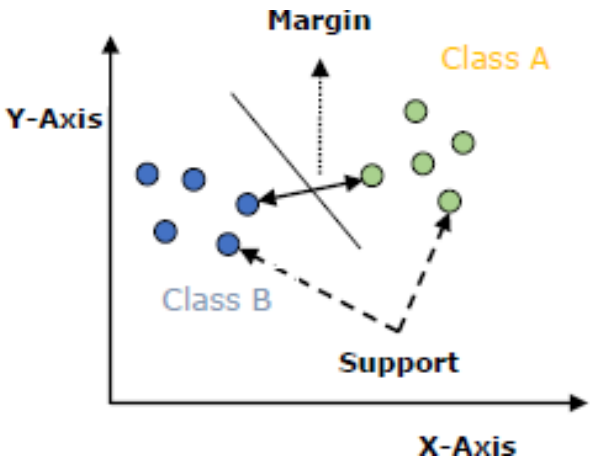



**Figure 2.3.2:** Support Vector machine outlay.

### *2.8.4. MATLAB Interface:*

MatlabR2020a for Machine learning includes the subsequent features:

- Clicking on applications for training and comparing models
- Advanced feature of extraction techniques
- Hyperparameter tuning on automatic premises and having selection to optimize model performance
- The ability to use the identical code for scaling the processing to big data and clusters
- Automated generation of C/C++ code for embedded and high-performance applications
- Known Popular classification, regression, and clustering algorithms for supervised and unsupervised learning
- Faster executing the data than open source on most statistical and machine learning computations.

# CHAPTER 3

# MATERIALS AND METHODS

## 3.1 Essentiality of Potential Drug target:

Tuberculosis remains a concern topic worldwide as the estimates say that it is among top 10 of the infectious disease and also above HIV/AIDS. Around 1.5 million people died in recent 2018 leading to the most prevailing infectious disease. Tuberculosis is curable and preventable but difficult to diagnose early. Apart from diagnosis there is the main problem of the rise of multidrug resistant strains (MDR-TB) and extensively drug resistant tuberculosis (XDR-TB) which is a crisis to health sectors. It is one of the serious health targets to end this pandemic by 2030.

Tuberculosis is principally caused by *Mycobacterium tuberculosis* which may be a small, aerobic and non-motile bacillus. The bacteria mainly affect the respiratory passage in the human system and can also affect other parts of the body. The symptoms are shown earlier or at the later stage of the infection, based on this it is classified into latent and active. The onset of the disease leads to the latent phase and later progresses to the active form and it may lead to fatality if left untreated. This type of disease is spread through air and may cause symptoms as of cough and sneezing. The patients with HIV/AIDS or who smoke have higher tendencies to develop active TB.

The *Mycobacterium* targets the airway passage as the human systems are the reservoir of *M.tuberculosis* and it is easily phagocytosed by macrophages but they are unable to kill and digest the bacterium and as its cell membrane prevents the fusion of phagosome with lysosome and it provides antibacterial factors so its main pathogenesis could be a target of high lipid content which defines its clinical significance. It mainly blocks the early endosomal autoantigen1 (EEA1) and consequently the multiplication of bacteria within the host system remains unchecked within macrophages. It prevents acidification of phagosomes as it carries the UreC gene and also prevents maturation of phagosomes by the production of diterpene isotuberculosinol. The macrophage killing is evaded by bacterium by the neutralization of reactive nitrogen intermediates. Mtb specifically covers itself in 1-tuberculosinyladenosine (1-TbAd) which may be a special nucleoside and acts as an antacid and neutralize pH and induce swelling in lysosomes which is encoded by gene Rv3378c. It has a cord factor which retains its survival in host. Cord factor is basically a glycolipid molecule present in the cell wall of the *Mycobacterium*. Mtb has the genome H37Rv in which 250 genes are involved in fatty acid metabolism. From which one of the gene *panC* is taken into account in the project entitled. *panC* encodes the protein Pantothenate synthetase which is an essential precursor of Vitamin B5 leading to the biosynthesis of coenzyme A (CoA) and Acyl carrier protein (ACP) both playing major roles in energy and fatty acid metabolism. As these play essential roles in pathogenicity and growth of Mtb therefore it is an attractive target for antimicrobial drugs against tuberculosis.

The result of the increasing rate of multi-drug resistant strains of Mtb, there is a need of a novel target inhibitor to inhibit the function of tuberculosis without much side effects in the host system. In the case, **CoA pathway** is an attractive source for drug targets in varied bacterial

pathogens. Its uniqueness is that it is found in many living systems and also it has non-similar sequences between the higher and lower level organism's CoA biosynthetic enzymes.

## 3.2. Pantothenate Synthetase as a potential target:

The lipid rich cell wall of Mtb is an important component for its survival and pathogenicity. The cell wall contains vast number of genes which encodes enzymes essential in fatty acid metabolism which leads to biosynthesis of lipids providing primary source of energy for the bacteria during infection. Thus, CoA pathway is a potential discovery of new antibacterial drugs. One of the most prior genes is **panC** gene encoding the enzyme Pantothenate Synthetase which further leads to the production of Pantothenate (Vitamin B5 precursor) in Bacteria. Pantothenate biosynthesis then further leads to the production of CoA and Acyl carrier protein (ACP) which is an important component of fatty acid synthesis. It is mainly important for the survival of the bacteria but not to cause disease. PanC catalyzes the ultimate step in biosynthesis of pantothenate through a Bi Uni Uni Bi Ping pong kinetic mechanism which is additionally referred to as double displacement. [19] [20]

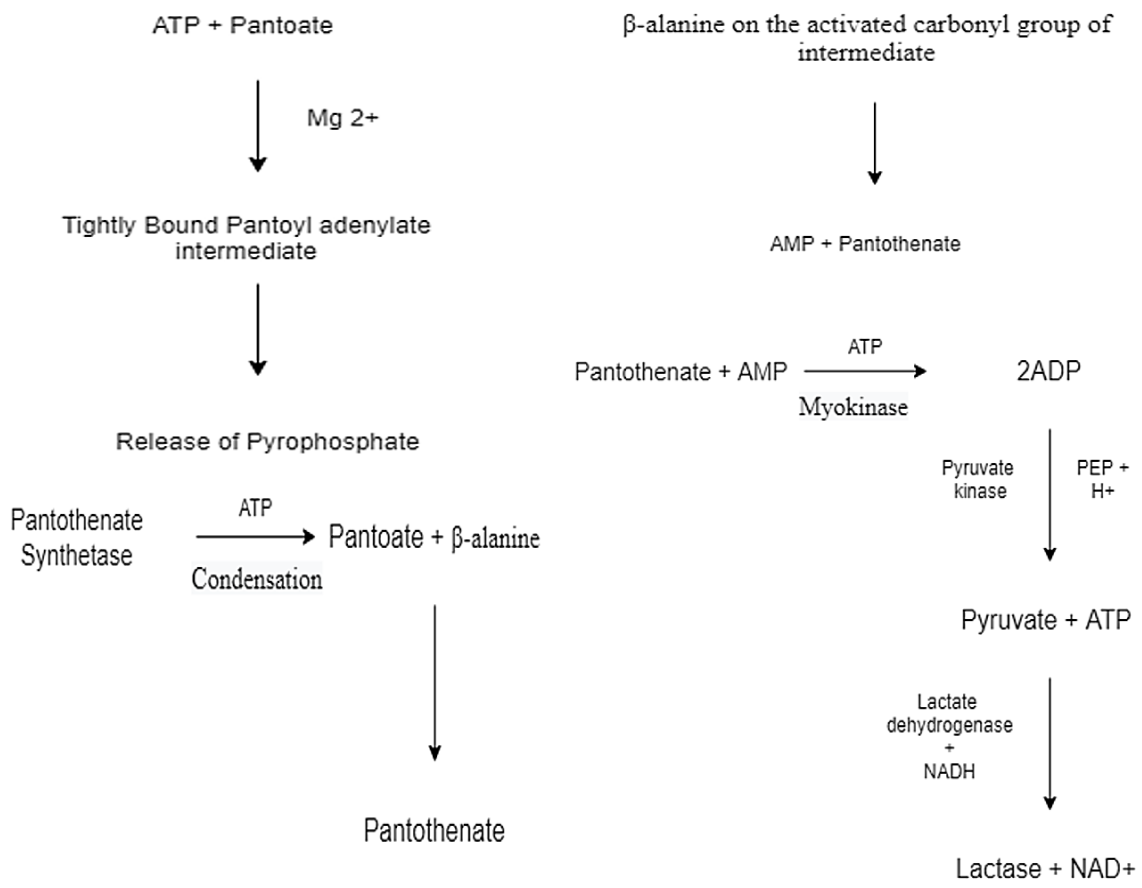



**Figure 3.1.** Reaction mechanism of Pantothenate Synthetase.

PanC being absent in humans they intake vitamin B5 through supplementary or through their diet. Pantothenate permease (PanF) is present in most of the bacteria but absent in Mycobacterium tuberculosis which scavenges pantothenate from the host when the bacteria have genetically disrupted PanC gene and becomes auxotroph requiring pantothenate supplementation for their growth. Therefore, we can say that there is no cross-reaction leading to toxicity in the host and thus makes panC an attractive drug target in Mtb.

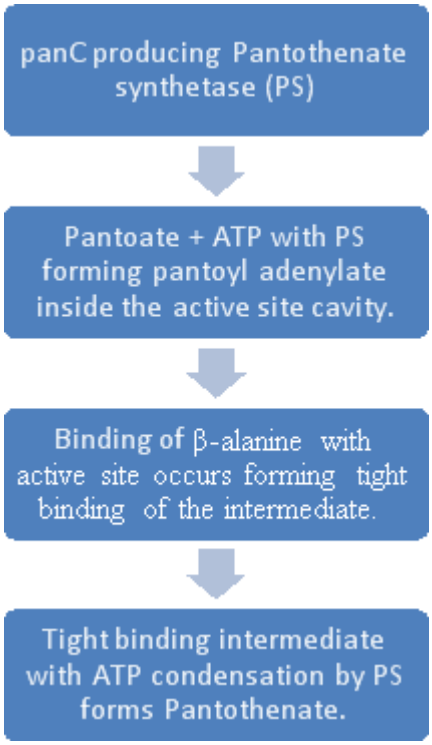



**Figure 3.2**. Flowchart describing the process of formation of Pantothenate.

## 3.3. Computational approach in finding the leads:

In Computational studies of targeting a protein with an inhibitor, the important step is the collection and refinement of the data. In this the protein was searched through uniprot which is a freely accessible protein database which also contains the functions of the proteins. The protein target was **Pantothenate synthetase (P9WIL5)** having a gene name of ***panC*** from the desired organism *Mycobacterium tuberculosis (strain ATCC 25618 / H37Rv)*. The Inhibitors against the protein was searched and filtered to get the desired results.

Computational approach is a very convenient step in finding the inhibitors of the protein by various Machine learning algorithms. As PanC is the desirable target for tuberculosis because of its absence in the mammalian system, so selected Pantothenate synthetase as a protein target and collected the data of its inhibitors through ChEMBL database which contains various bioactivity data of the desired protein. At last the docking process takes place to finally see the compatibility and binding of the result as inhibitor ligand with the protein.

This approach is beneficial and takes less time to predict the inhibitors.

## 3.4. Protein and Inhibitor Database:

The Universal Protein Resource database [46]is a freely accessible and comprehensive and central resource of protein sequences and functional annotation. It has four components for different purpose the UniprotKB was used which is an expertly curated protein database for our finding of the desired protein. The protein name was filtered out by the entry of "Antimicrobial resistance in Mycobacterium tuberculosis" and got 60 results out of which **Pantothenate synthetase (P9WIL5)** was found as an appropriate protein target.

The ChEMBL Database [47] is a biological database consisting of bioactive molecules with drug-like properties it is maintained by the European Bioinformatics Institute (EBI) of the European Molecular Biology Laboratory (EMBL). Through the entry of desired Protein target, obtained the various activity values and filtered through IC50 Values by the target of *Mycobacterium tuberculosis* and obtained about 126 values which was downloaded in csv format.

## 3.5. Data Pre-Processing:

After collecting the Data of the protein inhibitors, it is needed to be pre-processed further for the estimation of accurate result. In data pre-processing, the standard values of IC50, Molecule ChEMBL ID and SMILE structure were taken and Standard values were converted in the same unit in nM range. After conversion of the units, sorting method was applied from smallest to largest in which actives were taken in the range below 10,000 nm and then data was sorted in largest to smallest from which inactives were taken into account in the range above 500,000 nM. In this case inactives were less, so DUD- E (A database of useful decoys) was required to generate sufficient amount of data for inactives. Further the data were divided into two files of Actives and Inactives, and the duplicate values were removed from each file.

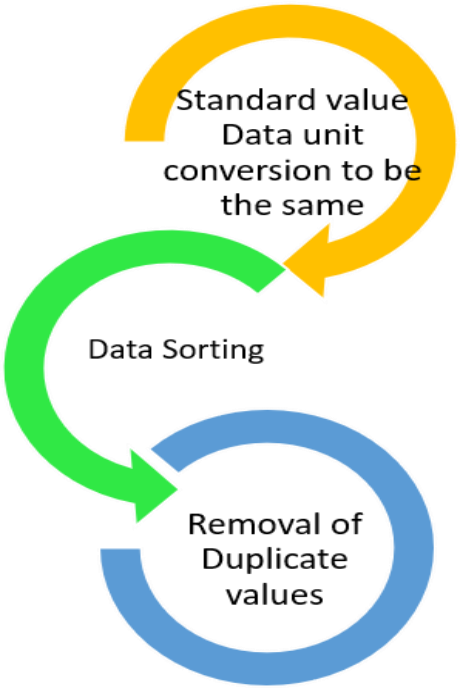



**Figure 3.3.** Data Pre-processing.

## 3.6. Descriptor Calculation:

After pre-processing the data, there is a need to calculate the property of chemical compounds so SMILE (Simplified molecular input line entry system) data was taken for descriptor calculation in smi format and PaDEL Descriptor software was used which is needed to calculate the molecular descriptors and fingerprints. The data was generated on the basis of MACCS fingerprint in csv format. MACCS fingerprint which is a chemical fingerprint containing a list of binary values (0,1) which characterize a molecule and are 166-bit structural key descriptors.

## 3.7. Model Building:

The protein model was built by the data prepared of Actives and Inactives descriptors as Training and Test set. The data was imported in MATLAB [44] environment in csv format and then through classification learner the data was trained by running the SVM (Support Vector Machine) and got good accuracy in linear SVM. So, the model was built by gaining confusion matrix and getting values by calculating the estimate through Matthews Correlation Coefficient (MCC) equation.

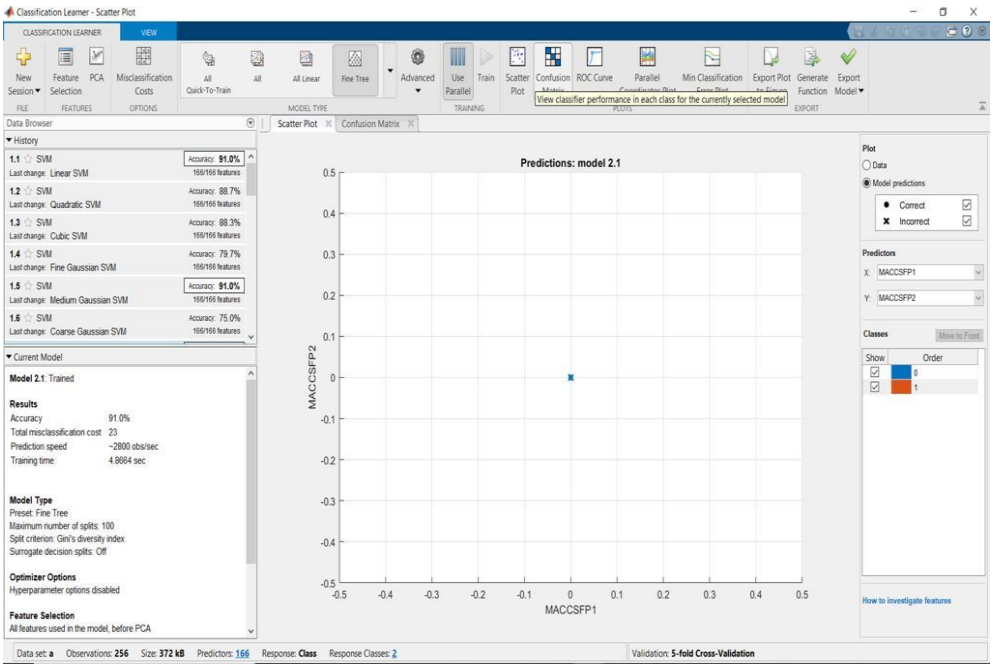



**Figure 3.4.** MATLAB Working Interface displaying the training of Protein Model.

## 3.8. Virtual Screening:

We have used Maybridge compound database for virtual screening selected 10 compounds from 55,717 compounds and prepared it for further docking process. The drugs in Maybridge library chemical space are diverse, drug-like and customarily follow Lipinski's rule of 5, are of prime quality and available in high stock quantities. The same process takes place as similar to making of compound database for model building.

## 3.9 Molecular docking:

The top 10 compounds as ligand structures were imported in **Schrodinger Maestro** [45] software and the ligand were prepared using **LigPrep**. Further docking was done through Glide and finally analysis of docked complex takes place in which the penalty and glide score together sums up to the result as Docking score. There are many types of forces occurring between the interaction of two molecules. The forces determine the docking score to predict how good is the interaction. The significant forces are: Van Der Waals, Electrical forces and hydrogen bonds.

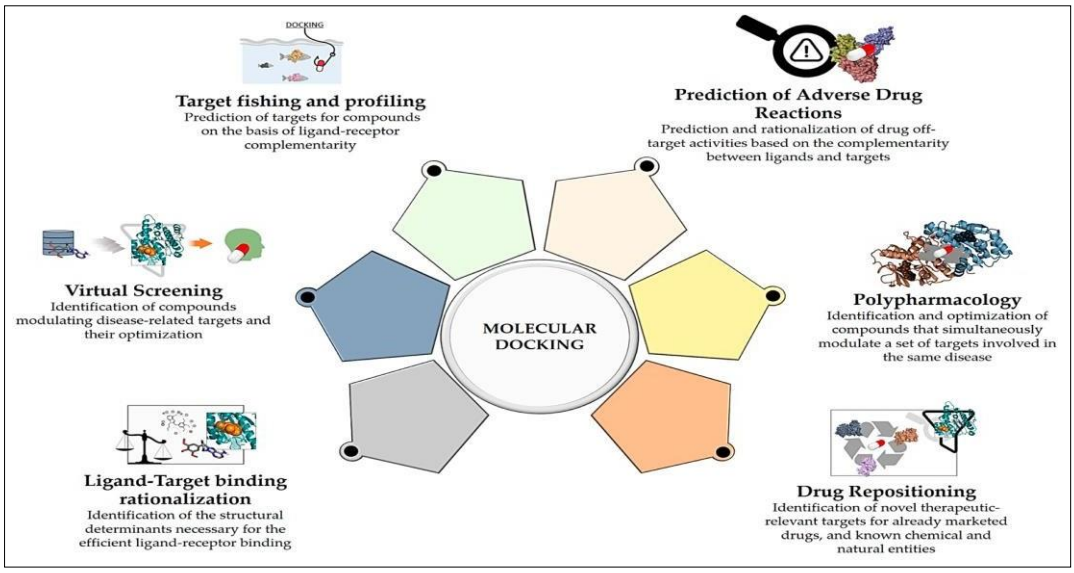



**Figure 3.5.** Molecular docking

## 3.10 Flowchart of the process:

**Figure 3.6.** Flowchart of the process.

# CHAPTER 4

# RESULT

## 4.1. Database collection and refinement:

The set of 126 inhibitors based on their IC50 values of *Pantothenate Synthetase* (P9WIL5) was collected for data to prepare. The data was refined based on active and inactive class, where the range of actives was lower than 10,000 nm and for inactives greater than 500,000 nm, so based on this 79 Actives and 3 Inactives were obtained. Due to the small dataset of inactives, explored DUD.E database [43] to get sufficient number of inactives. Inactives SMILE structure against one active SMILE structure was selected to be 240.

The active and inactive SMILE structure data were divided. The descriptors were calculated as MACCS fingerprint in binary form. For further Model building, classified the dataset into 80% Training set and 20% Test set respectively containing both actives (64,15) and inactives (192, 47).

## 4.2. Model Building:

The model was generated using Training and Test set in MATLAB environment. Imported the Training dataset and labelled it as 'a'. For classifying the data, trained under the Classification learner application in 5-fold cross validation to minimize overfitting. The Training dataset was trained in All- Quick- To- Train and 91% accuracy was obtained in Linear SVM. Matthews Correlation Coefficient (MCC) was calculated through Confusion matrix and MCC value as 0.76 was obtained. The trained model in MATLAB [44] workspace was imported by the label 'Trained Model" then imported the test dataset into the workspace and labelled it as 'b'. The True Positive, True Negative, False Positive and False Negative obtained for the test set as well and obtained the MCC value as 0.75.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Sensitivity (SE)} = \frac{TP}{TP + FP}$$

$$\text{Specificity (SP)} = \frac{TN}{TN + FP}$$

$$\text{MCC} = \frac{(TP * TN) - (FN * FP)}{\sqrt{(TP + FN) * (TN + FP) * (TP + FP) * (TN + FN)}}$$

| Model (SVM) | TP | TN | FN | FP | Sensitivity (SE) | Specificity (SP) | Accuracy (Q) | MCC |
|---|---|---|---|---|---|---|---|---|
| Training Set | 178 | 55 | 9 | 14 | 0.95 | 0.79 | 0.91 | 0.76 |
| Test Set | 13 | 46 | 2 | 4 | 0.86 | 0.92 | 0.90 | 0.75 |

Table 4.1. Model data

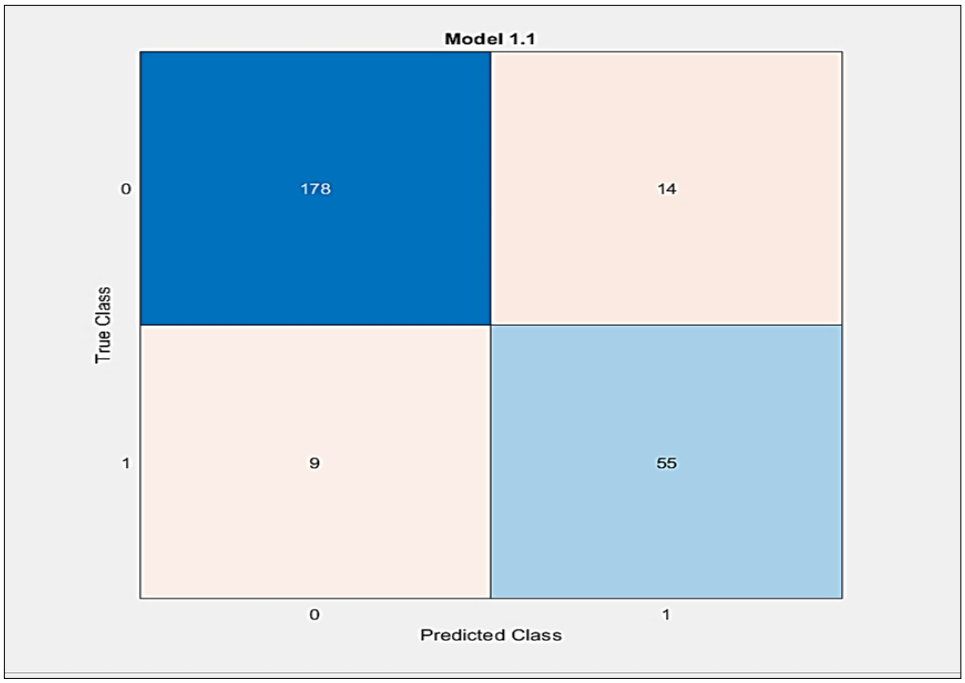



**Figure 4.1.** Confusion matrix of Training Model

## 4.3. Virtual Screening:

To validate the model generated, the virtual screening of approximately 55,717 compounds present in Maybridge library was performed. All database compounds were assigned in PaDEL software to 166 bits of MACCS fingerprint. 10 hits were returned with virtual screening and subjected to further docking studies.

## 4.3. Molecular Docking:

After getting a better result of the model further the control Nafronyl Oxalate was taken as ligand for the protein *Pantothenate Synthetase* (P9WIL5). The PDB file of protein was downloaded and visualized in UCSF Chimera software and further SDF file of Nafronyl Oxalate was downloaded from Pubchem accordingly and visualized. Removal of hydrogens from these molecules was done along with Torsion tree values set to default.

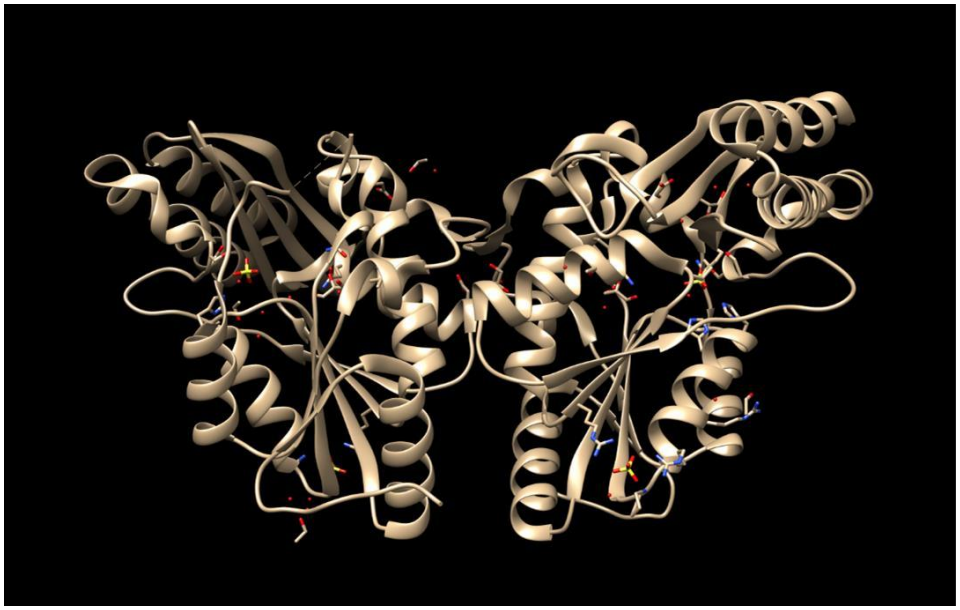


**Figure 4.2.** Protein Visualized in UCSF Chimera.

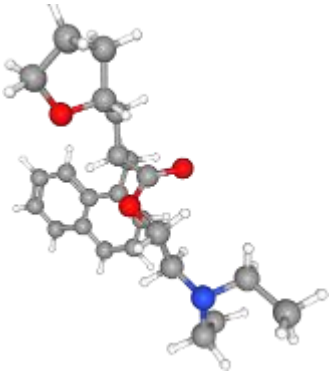


**Figure 4.3**. Nafronyl Oxalate

Prepared the docking complex of the control in Maestro Software and got Glide score of -6.160.

### 4.3.1. Ligand preparation:

Prepared the ligands from the Maybridge library and using LigPrep by first importing ligand data into the Project table from the library and then using the structure for further docking process. Applying Force field of OPLS_2005 which stands for Optimized potentials for liquid simulation. For Ionization, selected 7.0 +/- 2.0 and selected option "generate possible states at target pH using Epik" (Licensed version) and selected both the options "Desalt" and "Generate Tautomers." For Stereoisomers, selected "Determine chirality from 3D structure" and then selected the option of "Generate at most 32 per ligand" and low energy ring conformation to default 1 per ligand.

### 4.3.2 Grid Preparation:

After Ligand preparation, further Grid Preparation takes place. The grid from the known active sites were: His44 (H44), His47(H47), Asn69(N69), Glu72(Q72), Lys160(K160), Glu164(Q164). For grid preparation by default values were selected for 3D coordinates according to the active sites.

| Sr.no. | Compounds | Glide Score | Sites of Interaction (H-Bond) | Sites of Interaction (Salt Bridge) |
|---|---|---|---|---|
| 1) | Control (Nafronyl Oxalate) | -6.160 | His44, His47, Asn69, Glu72, Lys160, Glu164 | Asp161 |
| 2) | RJC02360 | -10.364 | Gly158, Gln164, Thr39, Gln72, Ser197, Hid44 | - |
| 3) | AW0555 | -9.580 | Ser197, Lys160, Gln72, Gln164 | Asp161 |
| 4) | AW00907 | -9.387 | Gln72, Lys160, Ser197 | Asp161 |
| 5) | NRB04426 | -9.257 | Val187, Lys160, Arg198, Gln128, Pro38 | - |
| 6) | NRB05197 | -8.772 | Lys160, Hid44, Ser197 | - |
| 7) | BTB12004 | -8.221 | Glu164, Asp161, Gly158, Glu72 | - |
| 8) | NRB04549 | -8.026 | Lys160, Asp161, Hid135, Glu72 | - |
| 9) | BTB13902 | -7.532 | Hid47, Arg198, Glu128 | - |
| 10) | BTB14725 | -7.059 | Ser196, Gly158, Lys160 | - |
| 11) | RJC01776 | -5.443 | Hid44, Gly158, Met40, Arg198, Glu128, Gly74, Asp78 | - |

**Table 4.2.** Ligand interaction and docking score of control and 10 compounds.

The results obtained as top 10 hits in which 9 (RJC02360, AW0555, AW00907, NRB04426, NRB05197, BTB12004, NRB04549, BTB13902, BTB14725) obtained good docking score were 1 (RJC01776) obtained poor docking score due to its value less than the docking score of the control complex.

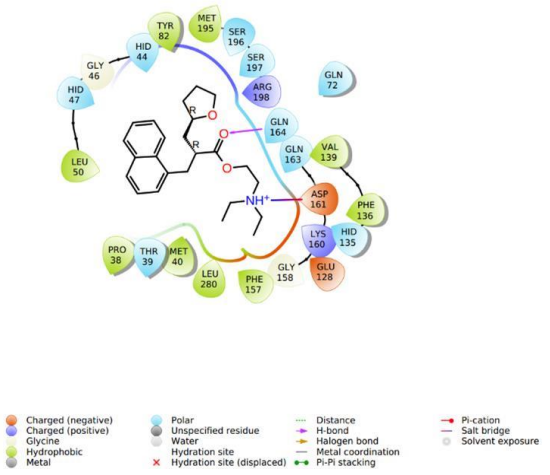
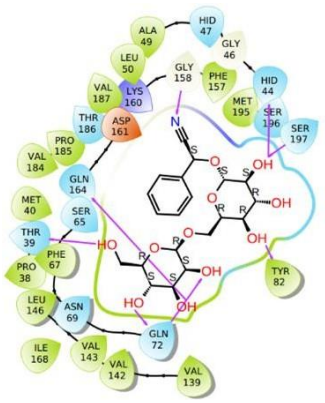
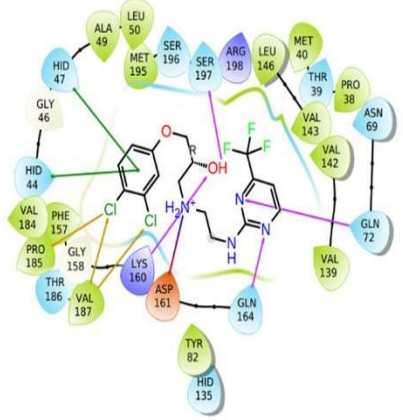


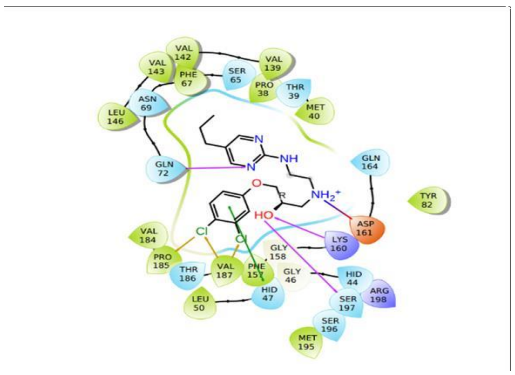

**Figure 4.4.** Ligand-Protein Interaction of control



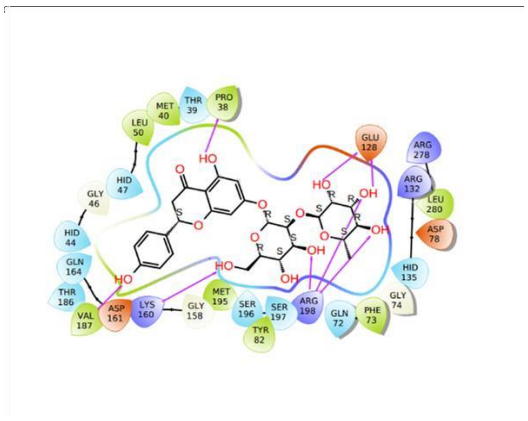
RJC02360



AW0555



AW00907



NRB04426

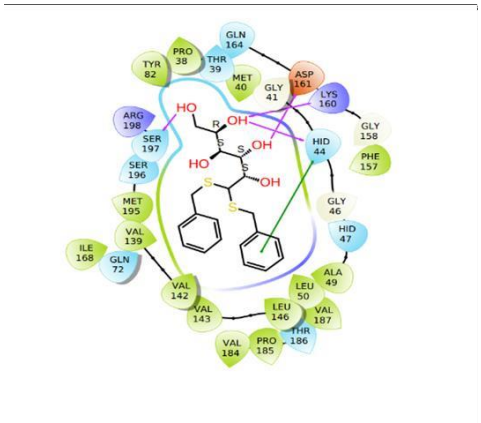
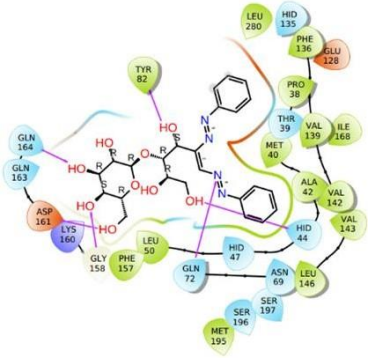
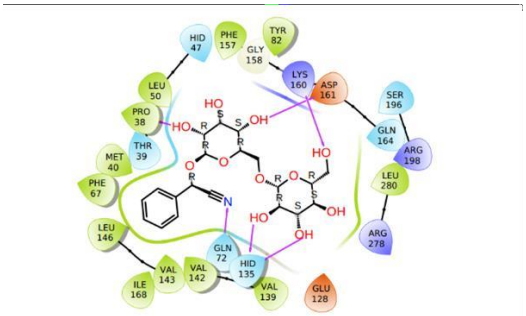
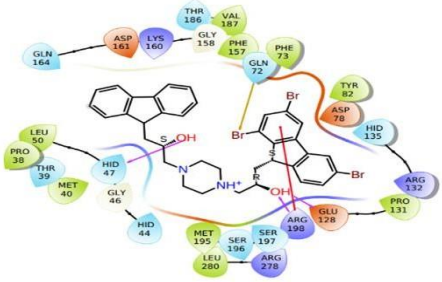
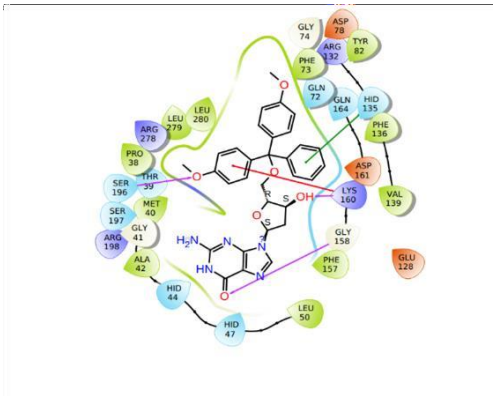
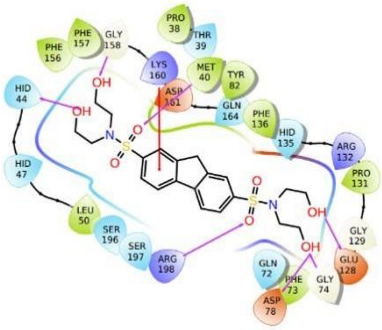
| | |
|---|---|
|  <br>NRB05197 |  <br>BTB12004 |
|  <br>NRB04549 |  <br>BTB13902 |
|  <br>BTB14725 |  <br>RJC01776 |

**Figure 4.5.** Ligand interaction of 10 compounds.

From the top 10 hits got the 9 compounds as potential inhibitors for the protein Pantothenate Synthetase.

## 4.4. Discussion:

By the computational approach we can validate the models of the inhibitor more precisely and in a time independent manner. For future review one can more accurately bring out the best model of the inhibitors of the protein *Pantothenate Synthetase*. In the project got the MCC value of 0.76 as it is a better prediction model but a best model can be made for further validation if possible. One can also use other classification technique apart from Linear SVM based on the accuracy. In the project, Structure based design model was built up, one can also synthesize the ligand molecule by De novo design of new ligands and also Optimization of the known ligands. This protein is a promising target for multi-drug resistant pathway as responsible for the survival of the mycobacterium.

## 4.5. Conclusion:

In the present research, we identified inhibitors for Pantothenate synthetase (P9WIL5) and prepared its data accordingly. Using Machine learning based classification modeling, a P9WIL5 inhibitor and non-inhibitor model was built. SVM model came out to be the best model. Further Virtual screening of the compounds was done from the ligands obtained from the Maybridge compound database and developed identified hits. The identified hits were further prepared for molecular docking in Schrodinger Maestro with the target protein. The compound RJC02360 came out to be the best potential inhibitor from the obtained top 10 hit compounds. This compound can be considered to be the inhibitor of the Pantothenate synthetase and can be further processed for validation.

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